CITY OF TORONTO 311 REQUESTS (2014-2025)

Insights into the data and forecasting monthly requests for service operations and resource planning

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CONTENTS

01 Project Summary

02 Modeling Workflow

03 What The Project Does Well

04 Environment Division Insights

05 Transport Division Insights

06 Seasonal Dependence Of 311 Requests

07 Recommendat ions

08 Next Steps

09 Appendix

PROJECT SUMMARY

Forecast 311 service requests across **Environment and Transportation** divisions using historical patterns, climate behavior, and ward-level distributions to support proactive operational planning.

Value to the City

- Enables **proactive scheduling** of salting, pothole repair, snow clearing, catch basin maintenance, wildlife response, and tree care.
- **Improves resource allocation** across the updated 25 ward system.
- **Reduces operational uncertainty** and prepares divisions for seasonal demand surges.
- **5-10% saving** of time, cost and efforts.

Takeaway

- Predictive forecasting offers a scalable, data-driven approach for shifting Toronto's operations from **reactive to proactive**, improving service levels and optimizing resource planning.

*For model predictions:
A2. Actual vs predicted (environment requests)
A4. Actual vs predicted (TRANSPORTATION requests)

MODELING WORKFLOW

Data Sources
- 311 Service Requests – Customer Initiated
- Green Areas, Tree Points
- Daily Weather Toronto

Feature Engineering
- Created **lagged request and weather features** (1-12 months) to capture historical behaviour.
- Generated **rolling window statistics** (2-3 months) for trend smoothening.
- Added **weather-derived predictors** such as freeze-thaw cycles, days since snow

ML Models
- Light GBM, XG Boost, Random Forest Regressor to predict next months requests (level of segregation: Wards, section)

Evaluation Metrics
- Average Error (MAE), Weighted Error (RMSE) and Variance Captured (R2 Score)

* Data Engineering Pipeline (Flow Chart)
A8. Analytical Pipeline Flow Chart

WHAT THE PROJECT DOES WELL

Harmonize Service Request Types

Standardizing request names, replacing outdated names with current ones ensures continuity and allows the model to learn long-term trends for each service request type

***A1. Harmonizing service request type (environment division)**

Harmonize Wards

Due to significant city restructuring during the study period, the project reclassified legacy 47 wards into new 25 ward system, using FSA-based matching to accurately map requests for outdated wards under new ones.

Ward-Level Monthly Splits

Predicts ward-level request totals and allocates them across service types using each ward's historical monthly distribution, ensuring realistic and seasonally accurate forecasts.

Mapping Hotspots

Creating hotspots for each service request type (where sufficient location data exists) enables the team to identify consistent areas of high activity and plan proactive inspections before peak request periods.

***A5. Road Damage/ Pothole Request Heat Map**

Reusability

Built scalable ML pipeline which can be used to analyze requests for other divisions. We can even remodel the pipeline to perform yearly or quarterly predictions which can help in planning city budget and allocating funds for maintenance operations.

ENVIRONMENT DIVISION INSIGHTS

Forestry Operations

- Dominates service requests with **88.59% share,** representing **331k total requests.**
- Top Request Type:
  - **General Pruning:** 175k total requests
  - **Tree Emergency Clean Up:** 93k total requests
  - Peak Month: **June – July**
- Top Ward:
  - **Etobicoke-Lakeshore:** 26k requests

Tree Protection and Plan Review

- Represents **37k (10%)** total requests focused on permit reviews and unauthorized activity monitoring.
- Top Requests Type:
  - **Unauthorized Tree Injury or Removal:** 16k total requests
  - **Dangerous Private Tree Investigations:** 15k total requests
  - Peak Month: **June – July**
- Top Ward:
  - **Etobicoke-Lakeshore:** 3k requests

Forestry & Natural Environment Management

- Specialized services in pest management and natural area maintenance.
- Top Request Types:
  - **Gypsy/Spongy Moth Inspections:** 2.3k total requests
  - **Natural Area Maintenance:** 1.5k total requests
  - Peak Month: **June – July**
- Top Ward:
  - **Etobicoke Centre:** 627 requests

*Predictions Dashboard:
A3. power bi dashboard (environment request predictions)

TRANSPORT DIVISION INSIGHTS

Road Operations

- Dominates service requests with **71.87% share**, representing **736k total requests.**
- Top Request Type:
  - **Road Pothole/ Road Damage:** 149k total requests
    - Peak Month: **Feb-May**
    - Top Ward: **Eglinton-Lawrence**
  - **Debris Clean Up:** 62k total requests
    - Peak Month: **April, May and November**
    - Top Ward: **Etobicoke-Lakeshore**

Turning Movement Counts (TMC)

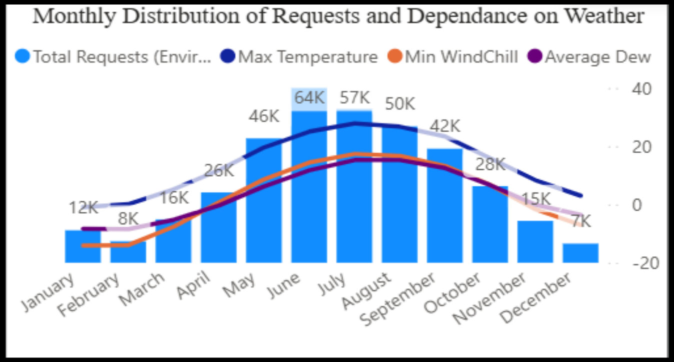
- Requests coverings traffic signs and signals accounts for **127k (13%) total requests**
- Top Request Type:
  - **Missing/Damaged Signs:** 57k total requests
    - Top Ward: **Spadina-Fort York**
  - **Traffic Signal Repair:** 15k total requests
    - Top Ward: **Etobicoke-Lakeshore**

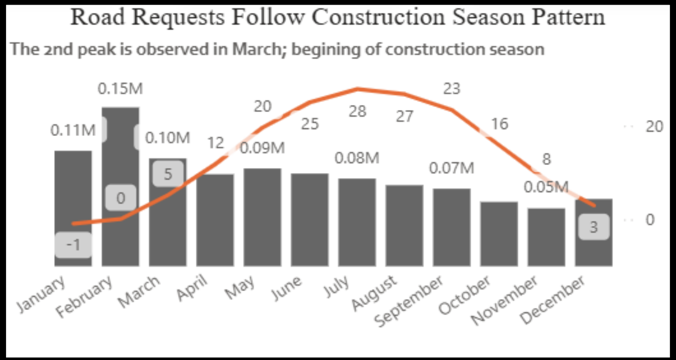
Right of Way (ROW)

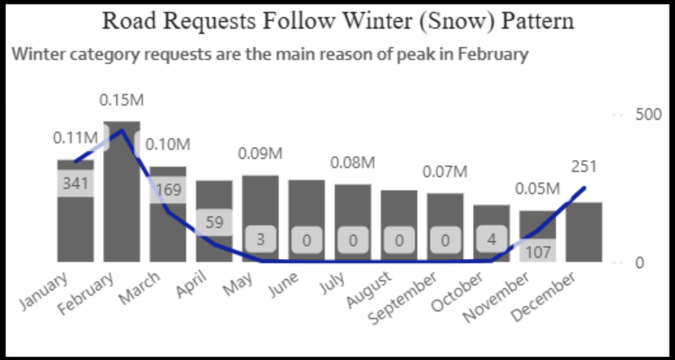
- Requests specializing in obstruction to the safe usage of public roads, contributes **10% of total requests.**
- Top Request Type:
  - **Report An Encroachment On City Property:** 45k total requests
    - Peak Month: **June – July**
    - Top Ward: **University-Rosedale**
  - **Illegal Snow Dumping:** 25k total requests
    - Peak Month: **Jan – Feb**
    - Top Ward: **Toronto-St. Paul's**

*Predictions Dashboard:
A5. power bi dashboard (Transportation requests predictions)

SEASONAL DEPENDENCE OF 311 REQUESTS







Environment Division requests, shows a **summer driven pattern,** with request volumes peaking between May and September.

Transportation Division requests are influenced by two seasonal patterns

Winter Maintenance Requests (remove snow, salt roads) are higher during **colder months.**

Road Damage Requests, increases in **March**; coinciding with **construction season** and the period with the **highest freeze-thaw cycles.**

RECOMMENDATIONS

Remove Outdated Service Request Types

The model continues to forecast requests for EAB (Emerald Ash Borer) exemption, even though the request type has been discontinued and shows no new activity after 2023. We can remove these requests from the model to maintain data accuracy of future predictions and reduce data segregations.

Improved Weather-Based Classifications

Re-categorizing existing service request types into weather-driven groups will enable the model to better capture seasonal variability, leading to higher predictive accuracy.

Incorporating Ward-Level Demographics

Several sections of 311 requests depend on the location, age of existing infrastructure as well as demographic of the reporting population. Integrating demographic and ward level characteristics will help the model reflect localized demand patterns accurately.

Incorporate policy changes

Policy changes can help us understand when one service request type ends (e.g recent law change to remove speed cameras will make street camera vandalism reports redundant). This will help us track and justify the changes we make to service level descriptions

NEXT STEPS

Validate pothole-prone locations by comparing them with collision hotspots, helping build a stronger case for **prioritizing repairs to reduce accident risk**.

Extend the forecasting framework to other city divisions by harmonizing service request types.

Create an app to present insights from this model to **support future quarterly or annual forecasts**, enabling the City to plan budgets, allocate crews, and prepare resources proactively—shifting maintenance operations from reactive to preventive.

APPENDIX AND SUPPORTING VISUALS

Original Service Requests	Starts	Ends	Renamed as	Comments
By Law Contravention Invest	2017	2024	Unauthorized Tree Injury or Removal	
Commercial Tree Storm Clean Up	2015	2024	Commercial Tree Emergency Clean Up	
Eab Exemption Request	2014	2023		
General Tree Inspection	2024	2024		1 Request Only
Gypsy Moth Control Insp	2016	2024	Spongy Moth Control Insp	
Hydro Brush Pick Up	2020	2024	Tree Brush Pick Up	
Ipm Inspection	2014	2024	Intergrated Pest Management Inspection	
Misc: Non-Forestry Activity	2023	2024		8 Requests
Mls Hazard Tree Invst	2014	2016		
Parks Ravine Safety Mtc Fnem	2014	2024	Natural Area Maintenance	
Permit Inspection	2014	2022		Divided into TTPR and RNFP Requests
Planting 11 Plus Trees Fnem	2014	2024	Planting 11 Plus Trees	
Private Tree Inspection	2014	2017	Dangerous Private Tree Inspection	
Ravine Inspection	2014	2022	Natural Area Maintenance	
Stemming	2014	2024	General Pruning	
Storm Clean Up	2014	2024	Tree Emergency Clean Up	
Stumping	2014	2024	Stump Removal	

A1. HARMONIZING SERVICE REQUEST TYPE (ENVIRONMENT DIVISION)

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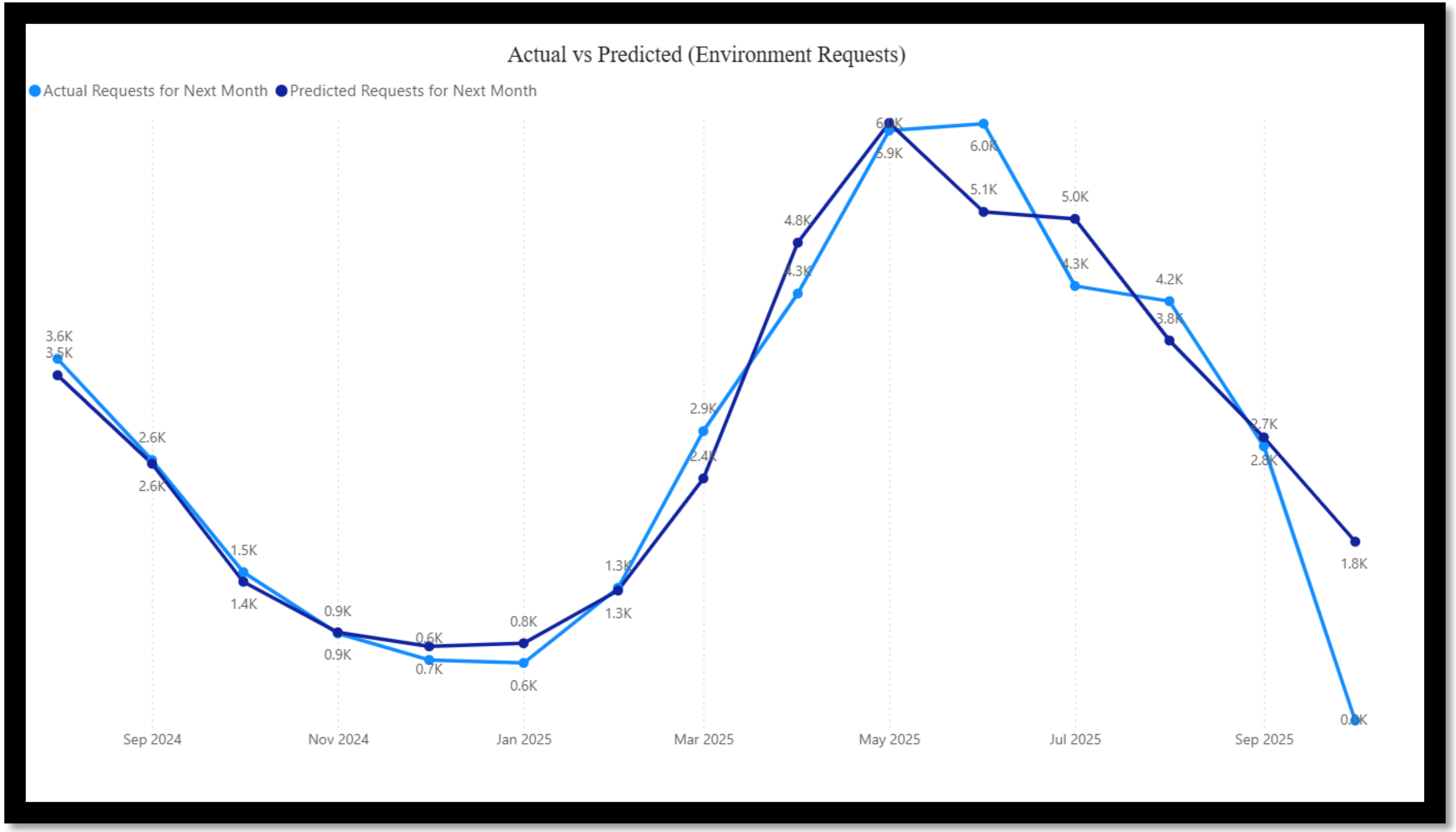
What the project does well



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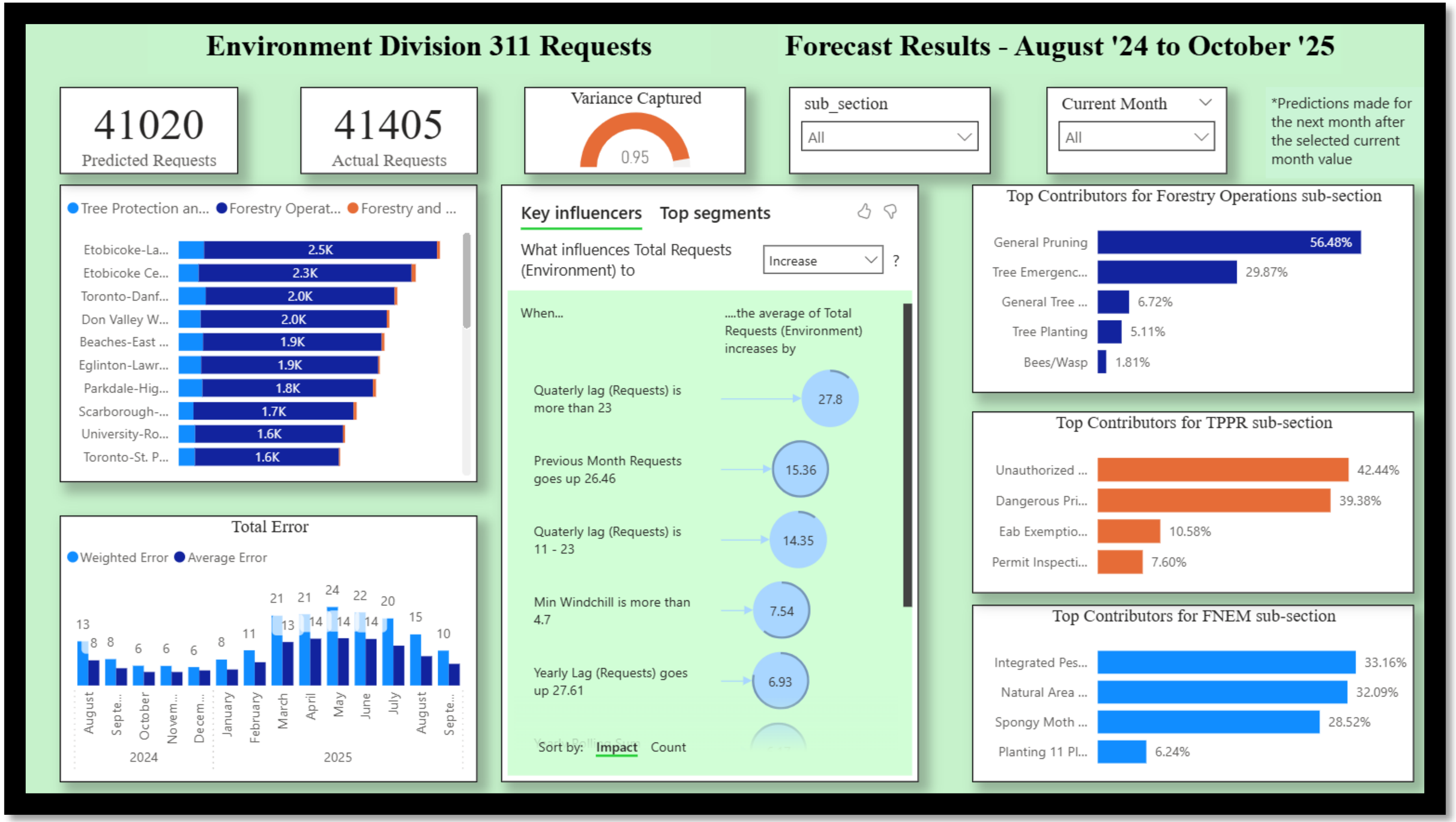
Project summary

A2. ACTUAL VS PREDICTED (ENVIRONMENT REQUESTS)



A3. POWER BI DASHBOARD (ENVIRONMENT REQUEST PREDICTIONS)

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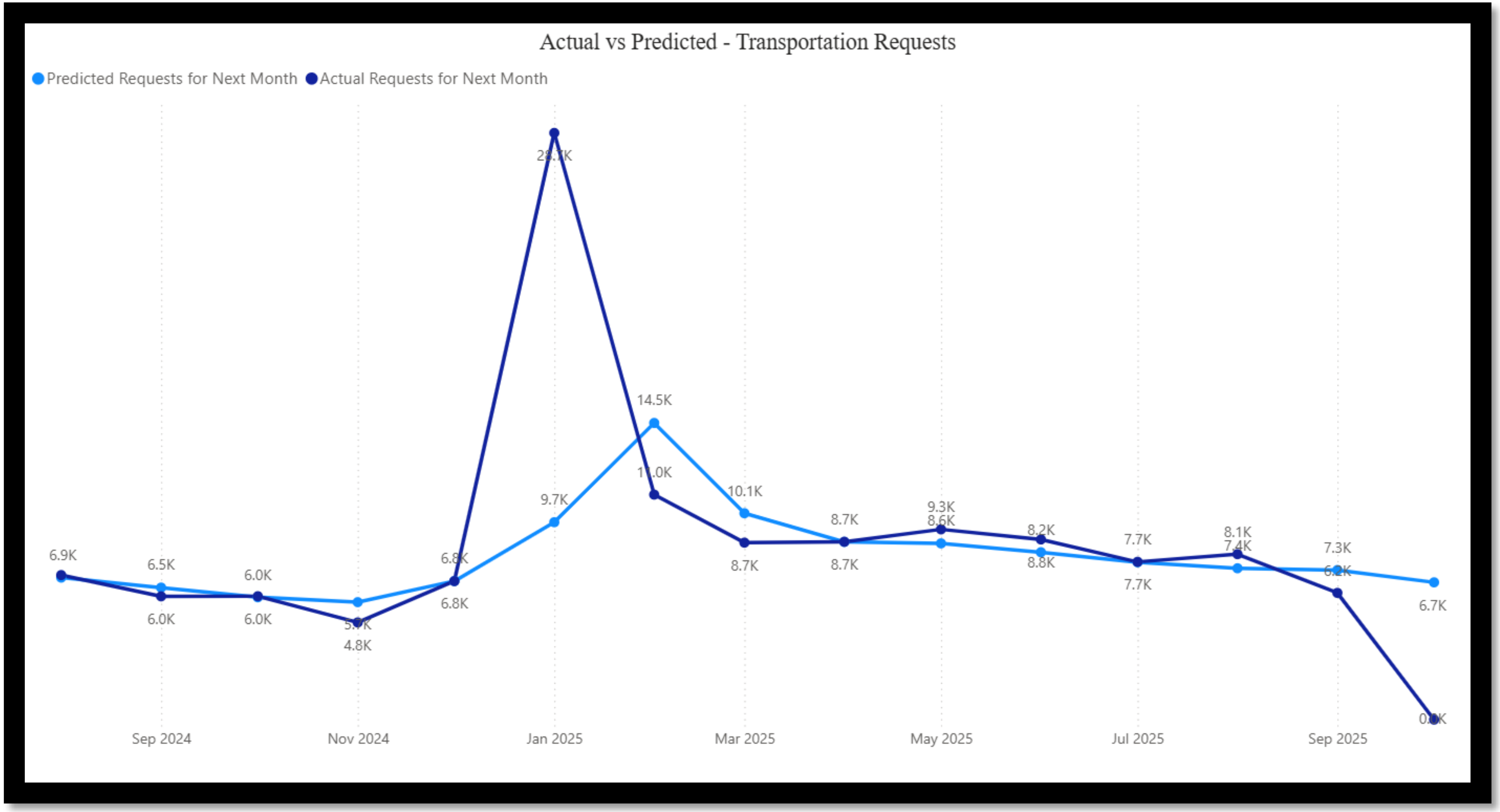
ENVIRONMENT DIVISION INSIGHTS



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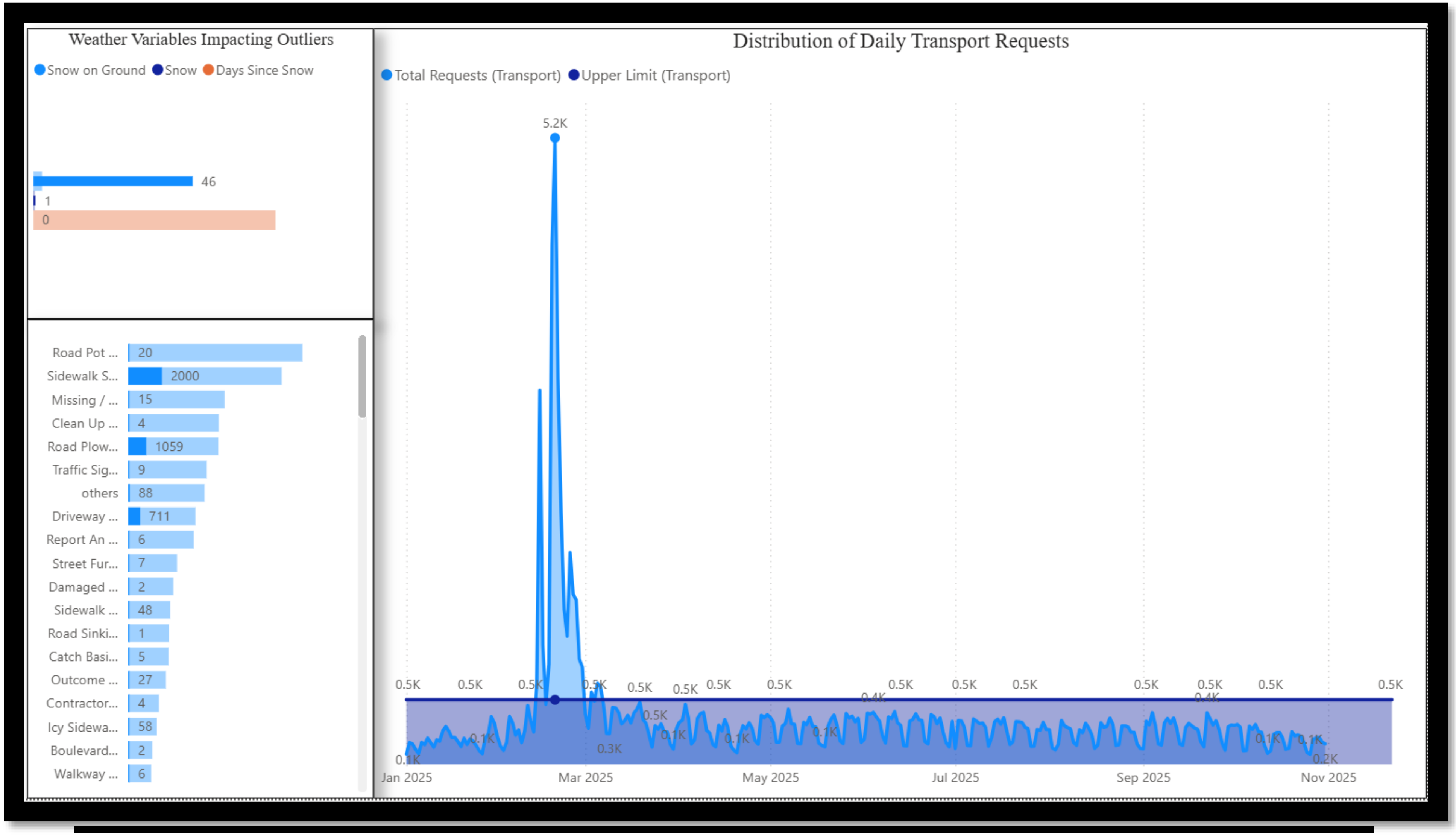
Project summary

* Explanation for large error in February prediction:

A5. extreme weather event (snow) contributed to higher wi...

A4. ACTUAL VS PREDICTED (TRANSPORTATION REQUESTS)



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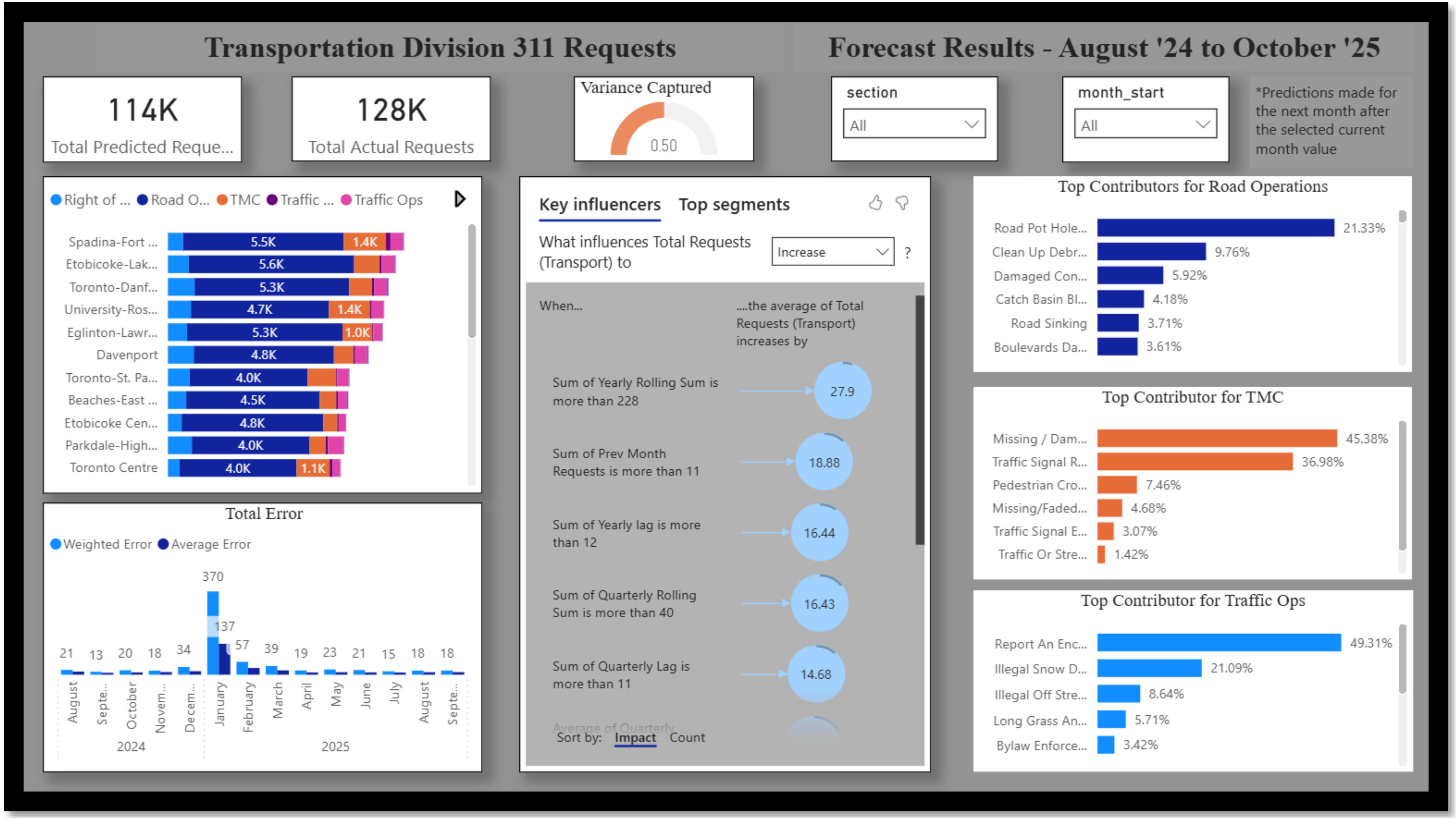
Project summary

A5. EXTREME WEATHER EVENT (SNOW) CONTRIBUTED TO HIGHER WINTER RELATED REQUESTS IN FEBRUARY 2025



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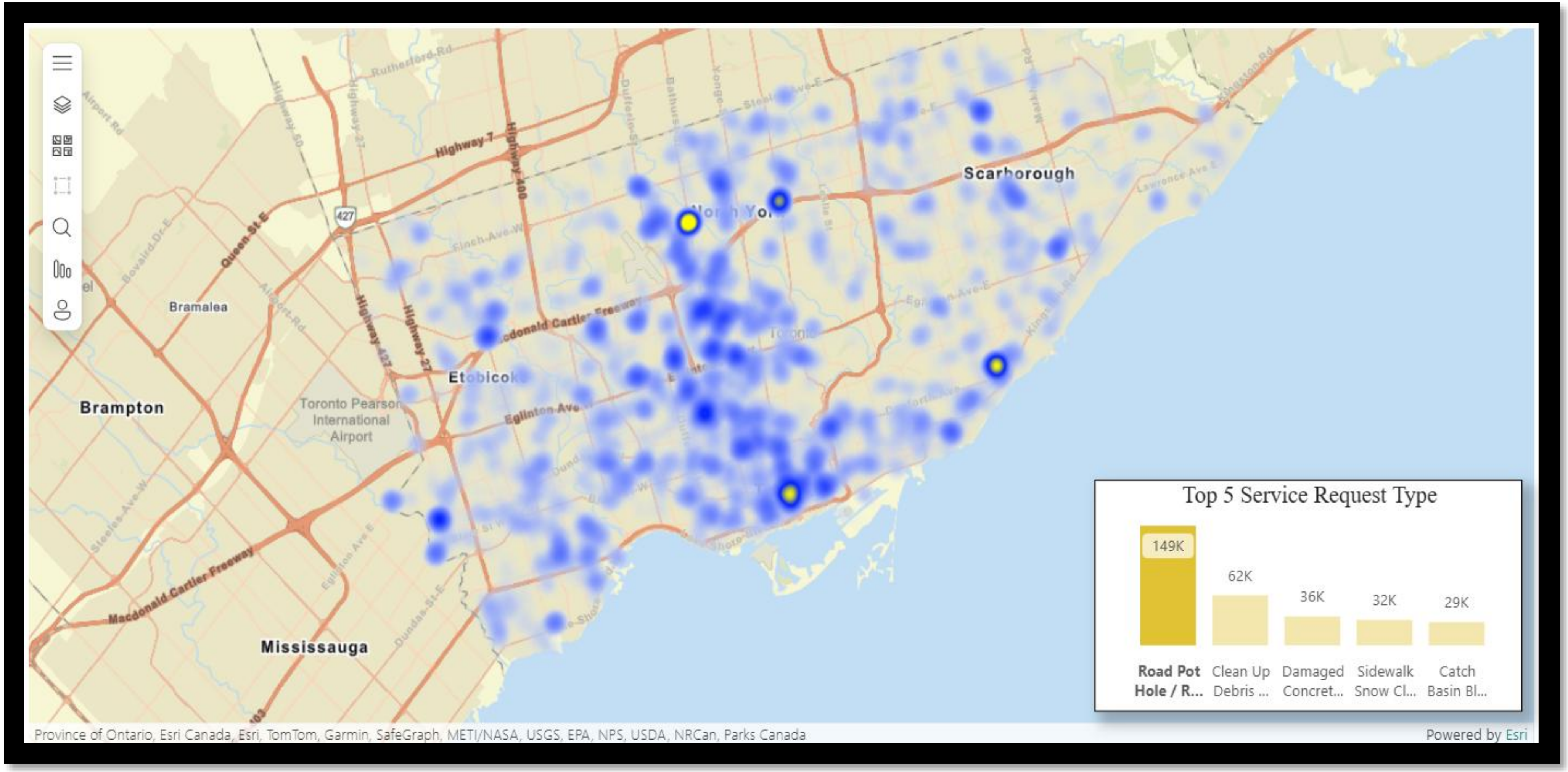
[Transport DIVISION insights]

A6. POWER BI DASHBOARD (TRANSPORTATION REQUESTS PREDICTIONS)



A7. ROAD DAMAGE/ POTHOLE REQUEST HEAT MAP

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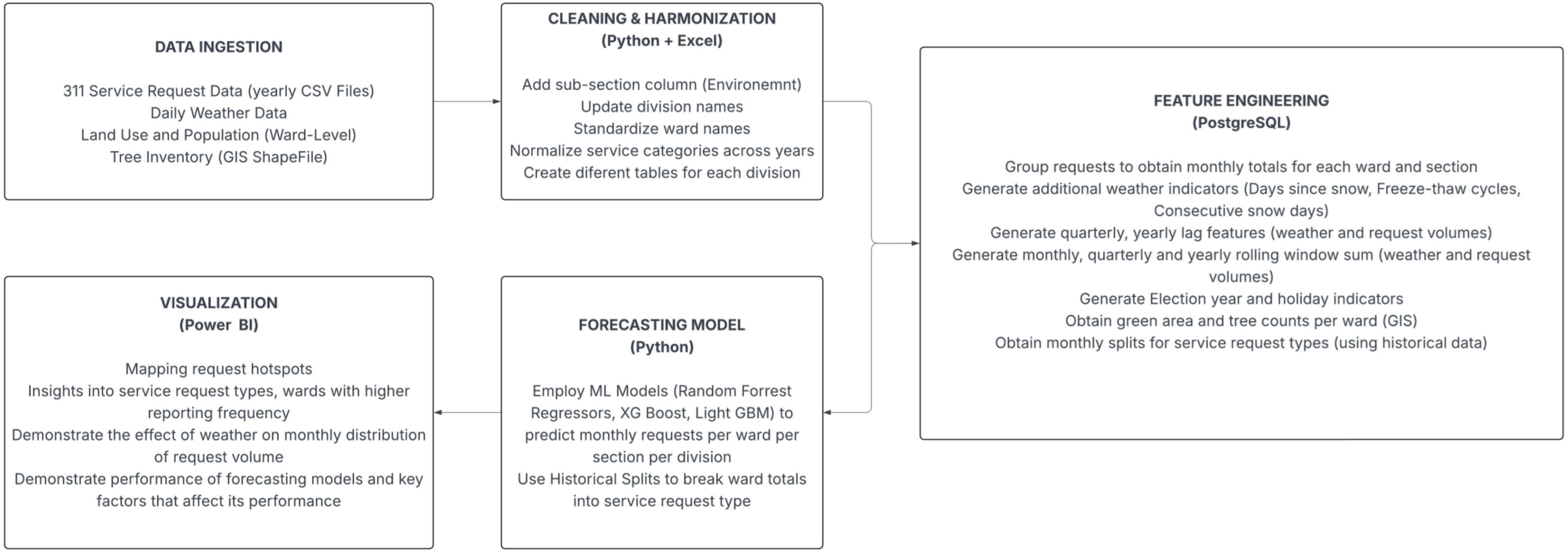
What the project does well

A8. ANALYTICAL PIPELINE FLOW CHART



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Modeling workflow

THANK YOU